

XIAOYING LIAO

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EDUCATION

Johns Hopkins University

2024–2025

Master of Science in Engineering in Biomedical Engineering

Baltimore, Maryland, USA

- Supervised by Prof. Carlo Iomini; GPA: 4.00/4.00
- Courses: Cellular Engineering, Tissue Engineering, Cell & Tissue Engineering Lab, Advanced Topics in Regenerative and Immune Engineering, Human Genetics, Systems Pharmacology and Personalized Medicine

University of Sydney

2020–2024

Bachelor of Science (Honours) in Statistics and Economics

Sydney, Australia

- Supervised by Prof. Jean Yang; GPA: 3.90/4.00; Rank: 1/200
- Courses: Probability and Estimation Theory, Statistical Inference, Stochastic Process, Advanced Statistical Modelling, Statistical Machine Learning, Life and Evolution

RESEARCH EXPERIENCE

Stanford University

2026–Present

Research Assistant

Stanford, California, USA

- Working with Prof. Xiaojie Qiu on single-cell and spatial omics to investigate cell-state transitions, tissue organization, and molecular programs underlying mammalian development and disease.
- Human fetal heart development: Analyzing PCW7 and 8 human fetal heart spatial transcriptomics data to characterize cardiac cell states, spatial niches, cell–cell communication, and 3D reconstruction during early heart development.
- 3D embryonic development: Building a 3D spatiotemporal atlas of mouse embryogenesis from E5.5–E7.5 to characterize cell-state transitions, spatial tissue organization, and developmental trajectories during early embryogenesis.
- Pediatric cardiomyopathy: Leading a single-cell multi-omics and spatial transcriptomics study comparing myocardial states before and after ventricular assist device support (VAD) and between recovery and non-recovery to identify cellular programs, spatial niches, and potential biomarkers associated with myocardial recovery.

Johns Hopkins University

2024–2026

Research Assistant

Baltimore, Maryland, USA

- Worked with Prof. Carlo Iomini at the Wilmer Eye Institute to investigate cilia-dependent Hedgehog signaling in meibomian gland development and dry eye disease.
- Meibomian gland development: Investigated how Hedgehog-responsive epithelial progenitors regulate gland development, meibocyte maturation, and lipid homeostasis.
- Cilia–Hedgehog signaling: Characterized Ift88–Smo genetic interactions and their effects on epithelial proliferation, gland morphology, and lipid production and composition.
- Single-cell transcriptomics: Generated and analyzed P5 and P10 meibomian gland scRNA-seq data to characterize developmental cell states and pathways associated with gland maturation and Hedgehog signaling.

University of Pittsburgh

2025–2026

Research Collaboration

Pittsburgh, Pennsylvania, USA

- Collaborated with Dr. Huanhuan Xiao and Prof. Liyun Zhang on quantitative visual behavior analysis in zebrafish models of inherited retinal degeneration.
- Visual behavior analysis: Analyzed photopic and scotopic visual motor response (VMR) data in pde6b mutant zebrafish to characterize genotype-dependent deficits in visual function.
- Quantitative analysis: Developed R-based workflows for behavioral data processing, response quantification, and statistical comparison across genotypes and lighting conditions.
- Therapeutic evaluation: Analyzed VMR responses across treatment conditions, including RA, 17-AAG, and DHA, and generated publication figures to evaluate potential rescue of visual function in the pde6b retinal degeneration model.

- Worked with Prof. Jean Yang and Prof. Clara Grazian on computational method development and benchmarking for spatial transcriptomics.
- Data preprocessing: Used Python and R to preprocess spatial transcriptomics datasets, including quality control, normalization, feature selection, and dimensionality reduction using principal component analysis (PCA).
- Spatial domain detection: Applied and compared clustering approaches, including K-means and Seurat, alongside spatially informed methods such as BayesSpace and STitch3D, to identify tissue domains and characterize spatial organization.
- Method evaluation: Benchmarked clustering performance across simulated and real datasets, examining factors affecting method efficacy and identifying strengths and limitations to inform method selection in spatial transcriptomics research.

PUBLISHED PAPERS, PREPRINTS & POSTERS

1. Python Song and **Xiaoying Liao**. *ProactiveEngine: Proactive Learning with Lifelong Memory via World Models*. Empirical Methods in Natural Language Processing (EMNLP, Main Conference), 2026.
2. Miao Cui (Presenting), Cinlong Huang, **Xiaoying Liao**, Xiaoke Song, Stephanie Conley, Jesse Engreitz, and Xiaojie Qiu. *Spatio-temporal Transcriptomic Profiling of Serial Human Fetal Heart Sections Reveals Cellular Architecture and Functional Niches during development*. Stanford Genetics Department Retreat (poster), 2026.
3. Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, and Carlo Iomini. *Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for Meibomian Gland Dysfunction Therapeutics*. Investigative Ophthalmology & Visual Science, 2026; pp. 3883.
4. Nianping Liu, Xuanchi Chen, Miao Cui, **Xiaoying Liao**, Xiaoke Song, Sairam Pantham, Weize Xu, and Xiaojie Qiu. *Bio-Babel: Autonomous Cross-Language Reconstruction of Agent-Ready Computational Biology Ecosystems*. bioRxiv, 2026.
5. Tianyang Wang, Xi Xiao, and **Xiaoying Liao**. *Foundation Models for Radiology Report Generation*. One chapter in medical field's book, 2026; Under review.
6. Xi Xiao, Yunbei Zhang, Lin Zhao, Yiyang Liu, **Xiaoying Liao**, Zheda Mai, Xingjian Li, Xiao Wang, Hao Xu, Jihun Hamm, Xue Lin, Min Xu, Qifan Wang, Tianyang Wang, and Cheng Han. *Prompt-based Adaptation in Large-scale Vision Models: A Survey*. Transactions on Machine Learning Research, 2026.
7. Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, **Xiaoying Liao**, Yingrui Ji, Janet Wang, Jianyang Gu, Yingqiang Ge, Weijie Xu, Xi Fang, Xiang Xu, Tianchen Zhao, Youngeun Kim, Tianyang Wang, Jihun Hamm, Smita Krishnaswamy, Jun Huan, and Chandan K. Reddy. *Agent Harness Engineering: A Survey*. Under review at ACL, 2026.
8. Tianyang Wang, Xi Xiao, Gaoqi Chen, **Xiaoying Liao**, Guo Cheng, and Yingrui Ji. *Boosting Active Learning with Knowledge Transfer*. 2025 IEEE 37th International Conference on Tools with Artificial Intelligence, 2025; pp. 1124–1131.

MANUSCRIPTS IN PREPARATION

1. Huanhuan Xiao, Xiangqian Shi, Abi Cha, Emma Liu, Randi Marshall, **Xiaoying Liao**, Molly Voit, Uzoamaka Nwagbo, Frazer Matthews, James H. Thierer, Meera Saxena, Ximin Li, Tae-In Kam, Xiangrong Kong, Jeff S. Mumm, Liyun Zhang. *Novel zebrafish genetic models of pde6b mutants exhibit rod photoreceptor degeneration*. Manuscript completed · Planned submission to Communications Biology by October 2026.
2. Huanhuan Xiao, **Xiaoying Liao**, Céline Portal, Zhenzhen Pan, Eileen Chen, Igor A. Butovich, Carlo Iomini. *Genetic interaction between primary cilia and Hedgehog signaling regulates meibomian gland development*. Manuscript in preparation · Expected submission by December 2026.
3. Huanhuan Xiao, Liyun Zhang, Emma Liu, **Xiaoying Liao**, Shaojun Du. *Expression and functional analysis of Hspa8 in early zebrafish development*. Manuscript completed · Planned submission to Cells in 2026.

PROFESSIONAL EXPERIENCE

JD.com

Feb. 2022 – May 2022

Data Analyst Intern

Beijing, China

- Retail Dataset Management: Utilized Hive SQL for complicated retail sales data management, focusing on home appliances at JD.com. Implemented time-series forecasting and multivariate regression to improve inventory and promotion strategies support for millions of users.
- Consumer Behavior Analytics: Analyzed consumer behavior using advanced statistical modeling in R and Python to identify key purchasing patterns. Developed and implemented sales strategies tailored to customer segments, resulting in a 25% increase in sales performance compared to the previous quarter.
- Operational Analytics Platform Development: Designed and built an advanced analytics dashboard using Python, enabling the operations team to track daily sales data in real time. Facilitated the rapid development of targeted business strategies, which led to a quantifiable 60% improvement in operational efficiency.

ByteDance – TikTok

Dec. 2021 – Feb. 2022

Data Operation Intern

Beijing, China

- Product Optimization: Performed market research for CRM platforms, delivering bespoke solutions to meet diverse global sales and operational challenges.
- Data Empowerment: Monitored business performance through SQL and Python, automatically analyzed vital indicators, and provided timely feedback on potential problems. Established an interactive BI data dashboard to display key business indicators visually and support real-time decision-making.
- Brand Growth: Conducted data analysis of user behavior to develop an AIPL (Awareness, Interest, Preference, and Loyalty) user classification model. Tailored payment tools, strategies, and creative content to users at various stages of brand awareness, achieving a remarkable 15% increase in brand advertising click-through rate.

TEACHING EXPERIENCE

Johns Hopkins University

2025–2026

Teaching Assistant

Baltimore, Maryland, USA

- Cell & Tissue Engineering Lab: Mentored students in practical modules on cell culture, gene delivery, metabolic glyco-engineering, and tissue modeling. Provided support in data collection, data analysis, and scientific lab reports.

University of Sydney

2023–2024

Teaching Assistant

Sydney, Australia

- STAT2011 Probability and Estimation Theory: Guided students through probability theories and statistical distributions, including practical R programming for data analysis applications.
- DATA2002 Data Analytics: Learning from Data: Led workshops on data analytics techniques, enabling students to apply statistical programming to real-world data challenges and machine learning concepts for informed decision-making.

SKILLS

Programming & Data Analysis: Python, R, SQL, MATLAB, Linux, Git, PyTorch, scikit-learn, NumPy, pandas, Scanpy, Seurat, Bioconductor, Cell Ranger, GraphPad Prism, FIJI/ImageJ

Computational Biology: Single-cell and spatial transcriptomics, Statistical modeling, Machine learning and deep learning, Foundation models for biological data, Multimodal and multi-omics integration, Perturbation and drug-response modeling, Spatial omics analysis and benchmarking

Experimental Biology: 10x Genomics single-cell sequencing, Illumina spatial transcriptomics, Cell and tissue culture, Immunostaining, Confocal and two-photon microscopy, PCR/qRT-PCR, Western blotting, Genotyping, DNA/RNA extraction, Mouse procedures

HONORS, AWARDS & CERTIFICATIONS

Dalyell Scholar – University of Sydney · 2022

Vice Chancellor's Global Mobility Scholarship – University of Sydney · 2020

2nd Prize, Roland Berger Case Competition – Rank: 2/2000 · 2023

3rd Meituan Business Analysis Elite Competition Participation Certificate – 2023

3rd Prize, Huawei Financial Elite Challenge – Australia Region – Rank: 3/1000 · 2021

2nd Prize, China National High School Biology Olympiad – 2018

Certified Financial Risk Manager (FRM) – Part I and Part II · 2023

ACADEMIC SERVICE

Conference reviewer: AAAI (2025–2026); ICLR (2027).

CONFERENCE PARTICIPATION & POSTER PRESENTATIONS

Stanford Genetics Department Retreat (2026). September 17–18, 2026 · CuriOdyssey, San Mateo, CA.

Association for Research in Vision and Ophthalmology (ARVO) (2026). Primary Cilia–Hedgehog Signaling Regulates Meibomian Gland Progenitors and Lipid Homeostasis; May 6, 2026 · Denver, CO.

Johns Hopkins Cell Biology Symposium (2026). Hedgehog Signaling Maintains Meibomian Gland Progenitors and Lipid Homeostasis with Implications for MG Dysfunction Therapeutics; April 10, 2026 · Bloomberg Center, Washington, DC.

Annual Postdoctoral Research Meeting (2026). Primary Cilia–Hedgehog Signaling Regulates Meibomian Gland Progenitors and Lipid Homeostasis; June 5, 2026 · Tilghman Auditorium, East Baltimore Campus, Baltimore, MD.

Maryland Stem Cell Colloquium (2025). Ciliary signaling pathways represent a potential intervention point for treating evaporative dry eye disease; October 16–17, 2025 · Maryland.

Annual Wilmer Research Meeting (2025). Cilia-mediated signaling pathways offer a therapeutic avenue for evaporative dry eye disease; April 25, 2025 · Maryland.

Sydney Bioinformatics Research Symposium (SBRS) (2023). Participant · University of Sydney.

Single-Cell Plus – Data Science Challenges in Single-Cell Research (2023). Participant · University of Sydney.

Unlocking Single Cell Spatial Omics Analyses with Scdny (2023). Participant · University of Sydney.

ADDITIONAL PROJECTS

Validating *Gne* gene Knockout CHO-K1 Cell Line

Sep. 2024 – Dec. 2024

Cell & Tissue Engineering Lab Advanced Project Leader

Baltimore, USA

- **Validation of CRISPR-mediated gene knockout:** Confirmed the successful deletion of *Gne* exons 2 and 4 in CHO-K1 cells using PCR and Sanger sequencing, verifying intended one-base deletion/insertion and precise editing outcomes.
- **Functional knockout confirmation at the protein level:** Conducted Western blot to assess Gne protein expression; observed absence of Gne signal in knockout cells while GAPDH served as loading control, supporting transcriptional-level findings.

Quantitative Evaluation of Benchmarking in Single-Cell Analysis

Feb. 2023 – June 2023

Supported by the Sydney Precision Data Science Centre and the World Health Organization (WHO)

USYD, Sydney

- **Developed a structured framework for literature evaluation:** Designed a comprehensive questionnaire with over 70 questions to assess all aspects of single-cell benchmarking research. Conducted a broad review to evaluate single-cell research, screening 245 pertinent studies from 563, facilitating a broad perspective on the field.
- **Statistical Modeling and Analysis:** Implemented varied regression analyses to discern critical factors in single-cell benchmarking, endorsing the use of multiple linear regression for robust predictive insights.
- **Differentiate Research Categorization:** Applied logistic regression in categorizing research papers into benchmarking versus method development. Revealed through data analysis that more method comparisons, speed measurements, and lower impact factors characterize pure benchmarking research papers.

Type 2 Diabetes Risk Prediction and Food Intake Recommendations

Aug. 2022 – Dec. 2022

Capstone Project Leader, STAT3888 Statistical Machine Learning

USYD, Sydney

- **Feature Selection:** Calculated correlations between features and HbA1c (Hemoglobin A1c), then identified key variables and created new features better to represent lifestyle and dietary influences on HbA1c levels.
- **Regression Analysis:** Developed robust linear, ridge, and Lasso regression models to evaluate the influence on HbA1c indicator, enhancing disease risk prediction accuracy.
- **Decision Tree Modeling :** Selected the best decision tree model through detailed cross-validation and accuracy comparison, then identified high-risk individuals using selected variables and provided personalized nutritional advice and health interventions to reduce HbA1c levels and improve health.